

Groups and Lie Algebras: Exercise Sheet 4

1. Projection operators for non-trivial irreducible representations

Let (V, ρ) be a finite dimensional representation of a finite group G and V_i an irreducible representation of G with character $\chi_i(-)$. Show that the operator

$$\frac{\dim(V_i)}{|G|} \sum_{g \in G} \overline{\chi_i(g)} \rho(g)$$

is projection operator on the (largest) sub-representation of V which is isomorphic to a direct sum of copies of V_i .

Apply this formula to the action of $\mathbb{Z}_n$ on the vector space of functions from $\mathbb{Z}_n$ to $\mathbb{C}$ to recover the decomposition into Fourier modes.

2. Dimensions of irreducible representations of S_n

Start with the Frobenius formula for the characters of the symmetric group.

$$\chi_\lambda(C_i) = [\Delta(\mathbf{x}) \prod_{j=1}^n P_j(\mathbf{x})^{i_j}]_{l_1, \dots, l_j}$$

Notation are as in the lecture and as in Fulton-Harris: The conjugacy classes C_i are labelled by vectors determining the cycle shape, $l_j = \lambda_j + k - j$, k number of rows of the diagram. Derive the following formula for the dimension of the representation V_λ :

$$\dim V_\lambda = \frac{n!}{l_1! \dots l_k!} \prod_{i < j} (l_i - l_j)$$

3. Schur functors We can use the representation theory of the symmetric group S_n to produce new representations of arbitrary groups. Let G be a group and V a representation of G .

- a) S_n acts on $V^{\otimes n}$ by permuting the tensor factors. Show that this action commutes with the action of G . This means that S_n acts by maps of G representations or equivalently G acts by maps of S_n representations.
- b) We can decompose $V^{\otimes n}$ into the irreducible representation V_λ of S_n indexed by partitions of n

$$V^{\otimes n} \cong \bigoplus_{\lambda} S^\lambda(V) \otimes V_\lambda$$

Show that $S^\lambda(V)$ is a representation of G .

- c) The construction $V \mapsto S^\lambda(V)$ is called the *Schur functor* corresponding to the partition λ . Compute the Schur functor for $\lambda = (n)$ and $\lambda = (1, \dots, 1)$.
- $S^\lambda(V)$ is the image of the Young symmetrizer c_λ acting on $V^{\otimes n}$. Using this, compute $S^{(2,1)}(V)$ for a 2-dimensional representation V of G .